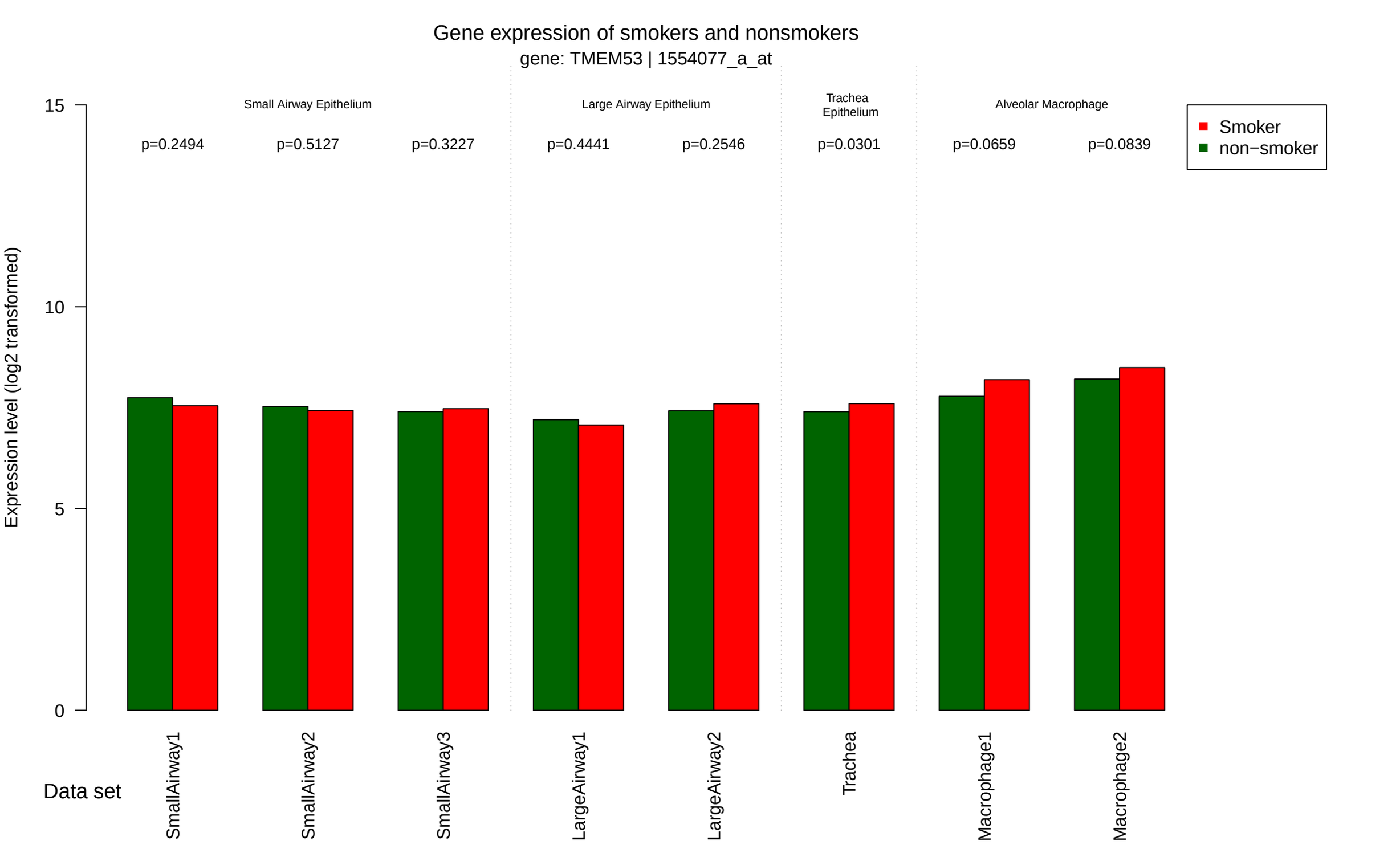
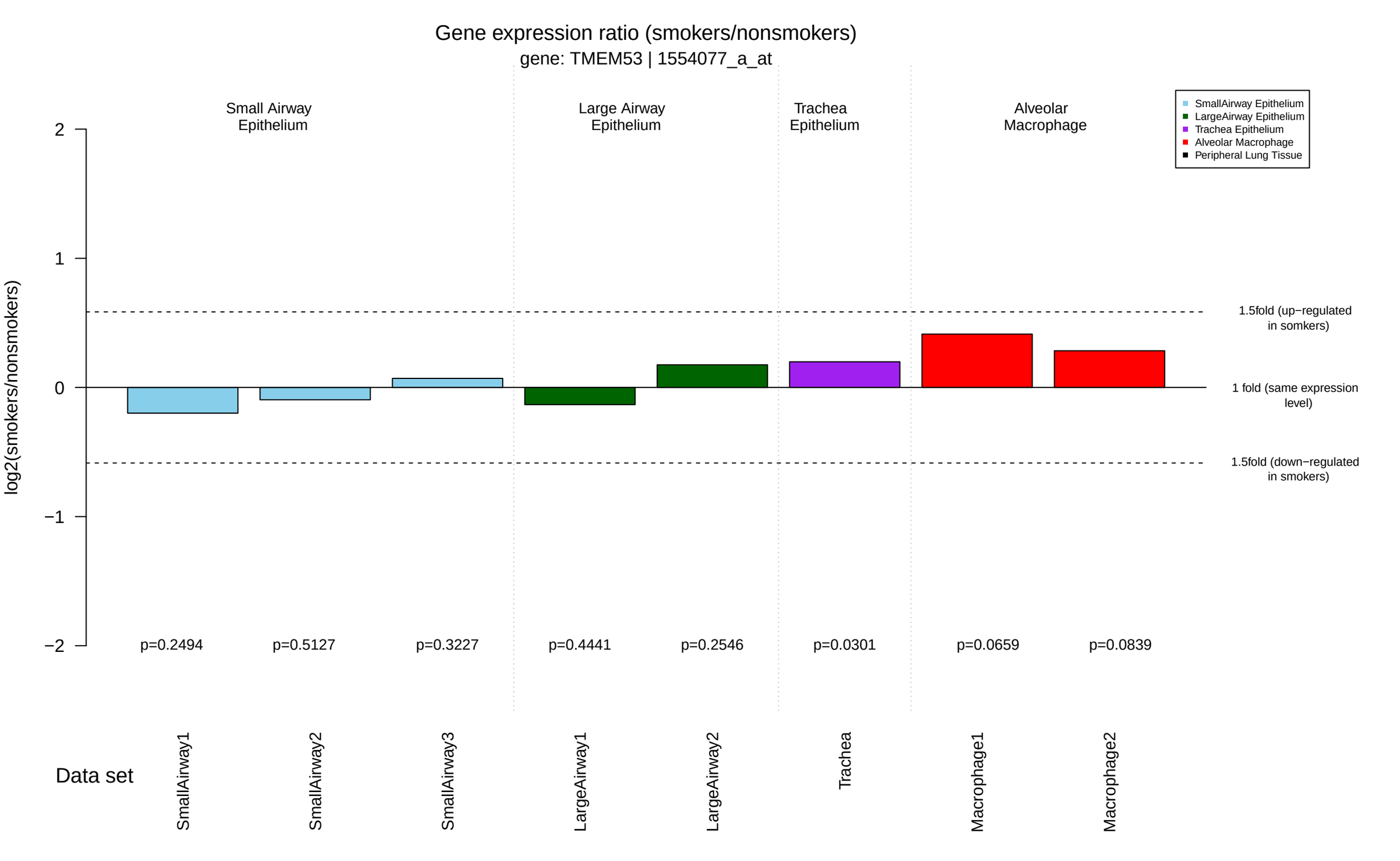
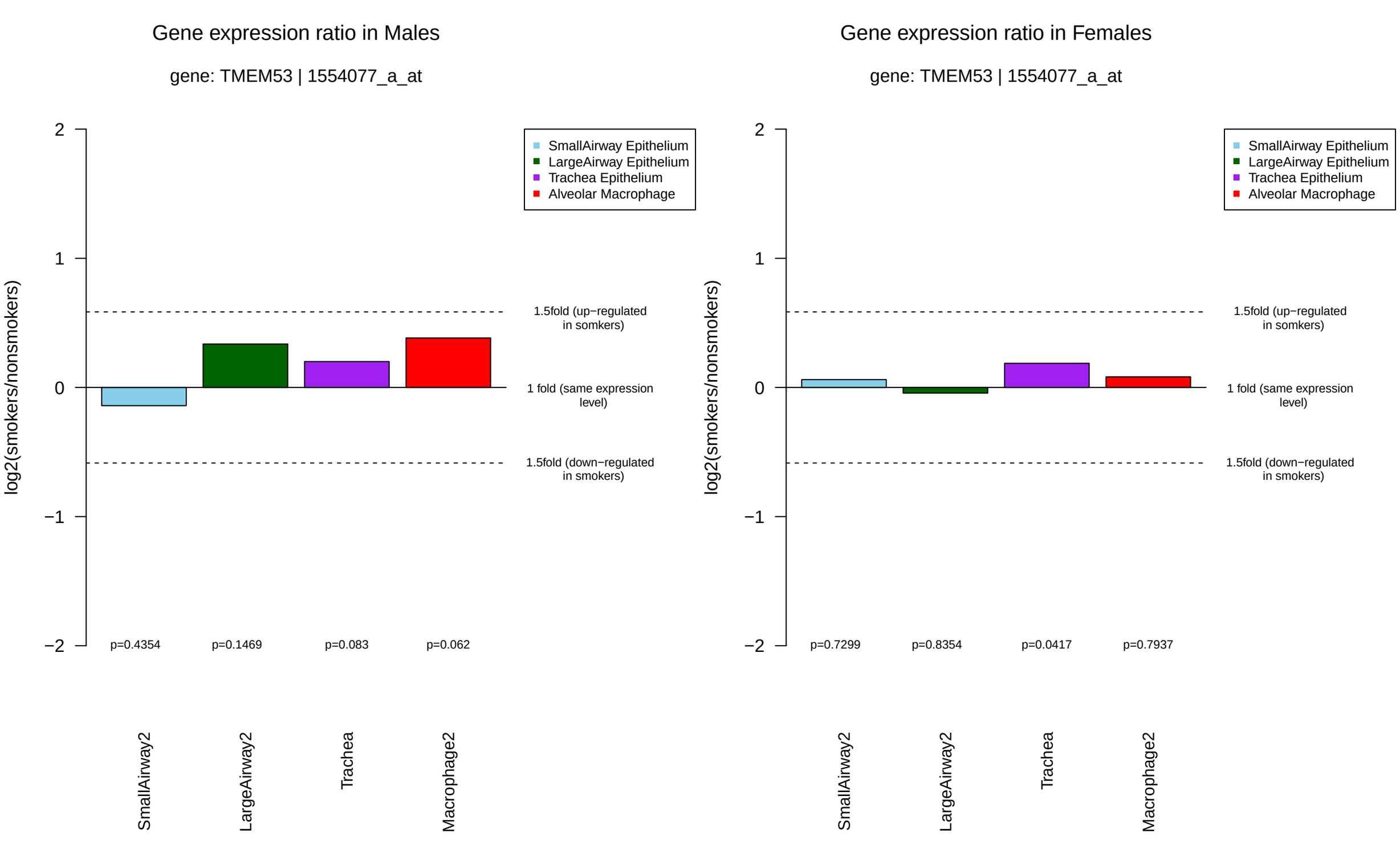
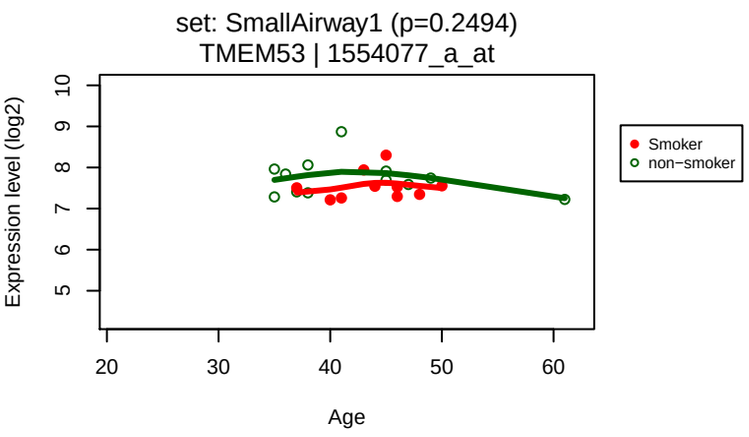
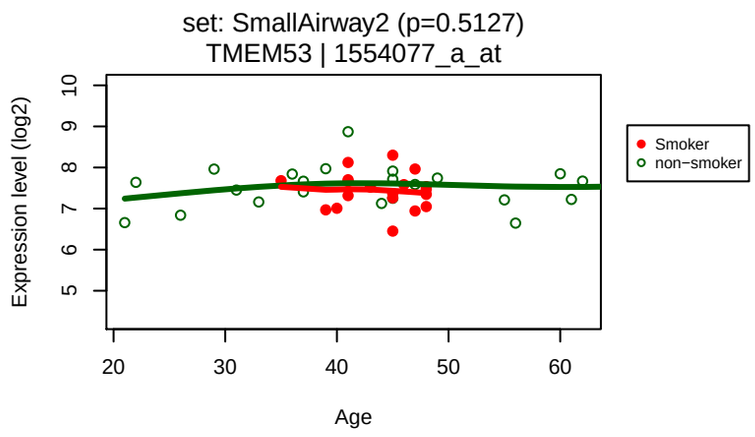
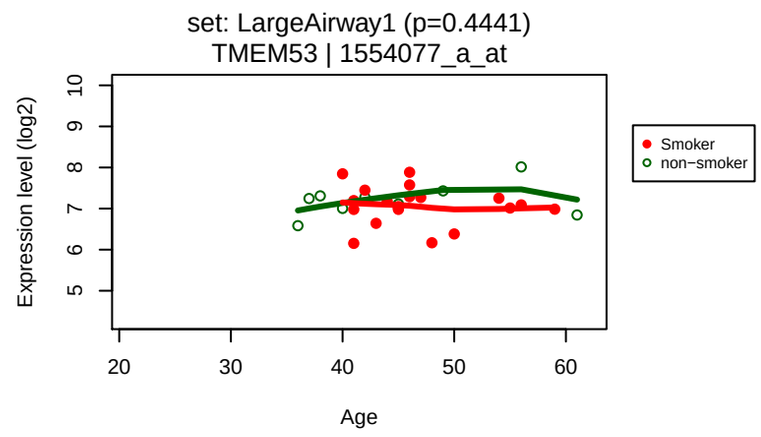

Gene expression of smokers and nonsmokers
gene: TMEM53 | 1554077_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2494
p=0.5127
p=0.3227
p=0.4441
p=0.2546
p=0.0301
p=0.0659
p=0.0839
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

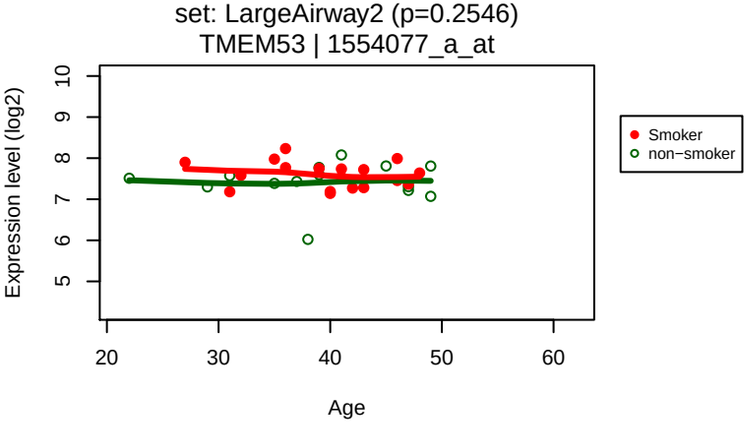

Gene expression ratio (smokers/nonsmokers)
gene: TMEM53 | 1554077_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2494
p=0.5127
p=0.3227
p=0.4441
p=0.2546
p=0.0301
p=0.0659
p=0.0839
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

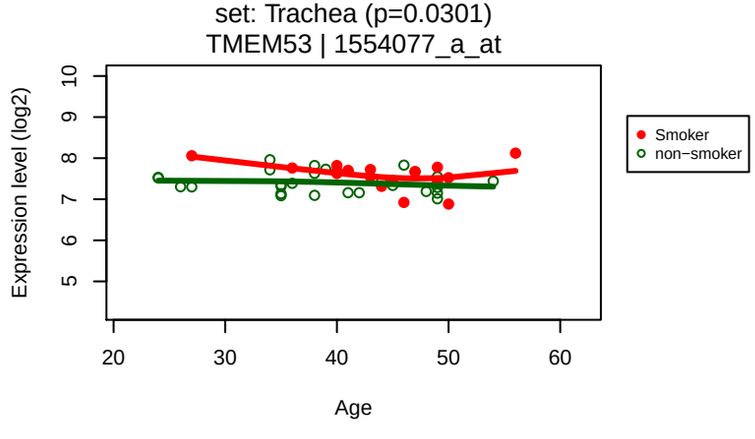

Gene expression ratio in Males
gene: TMEM53 | 1554077_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4354
p=0.1469
p=0.083
p=0.062
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM53 | 1554077_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7299
p=0.8354
p=0.0417
p=0.7937
SmallAirway2
LargeAirway2
Trachea
Macrophage2

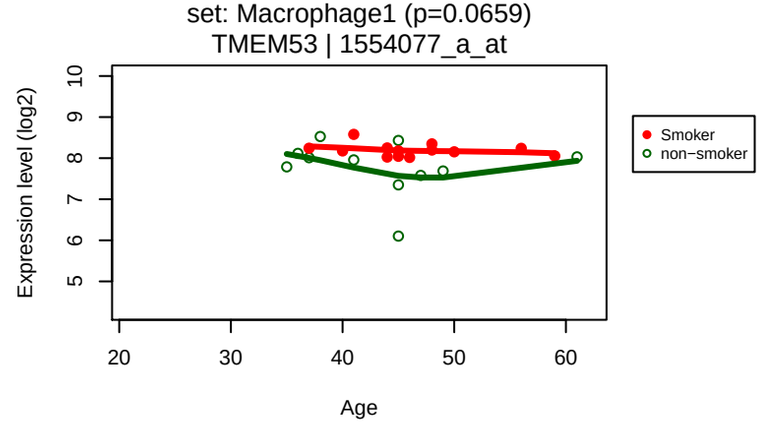

set: SmallAirway1 (p=0.2494)
TMEM53 | 1554077_a_at
Expression level (log2)
Age
Smoker
non−smoker

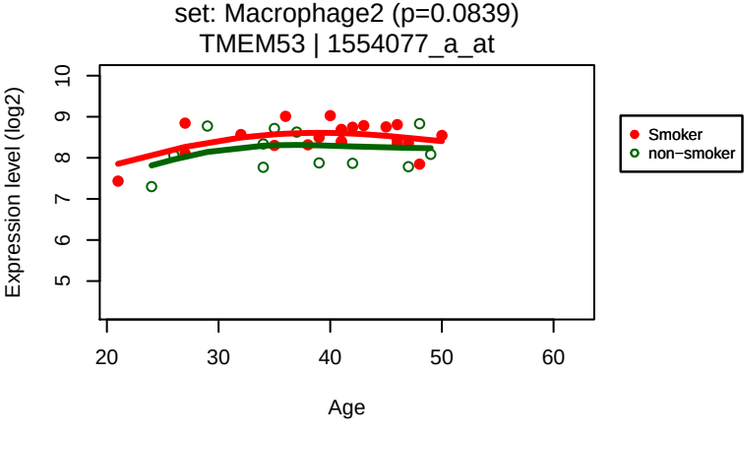

set: SmallAirway2 (p=0.5127)
TMEM53 | 1554077_a_at
Expression level (log2)
Age
Smoker
non−smoker

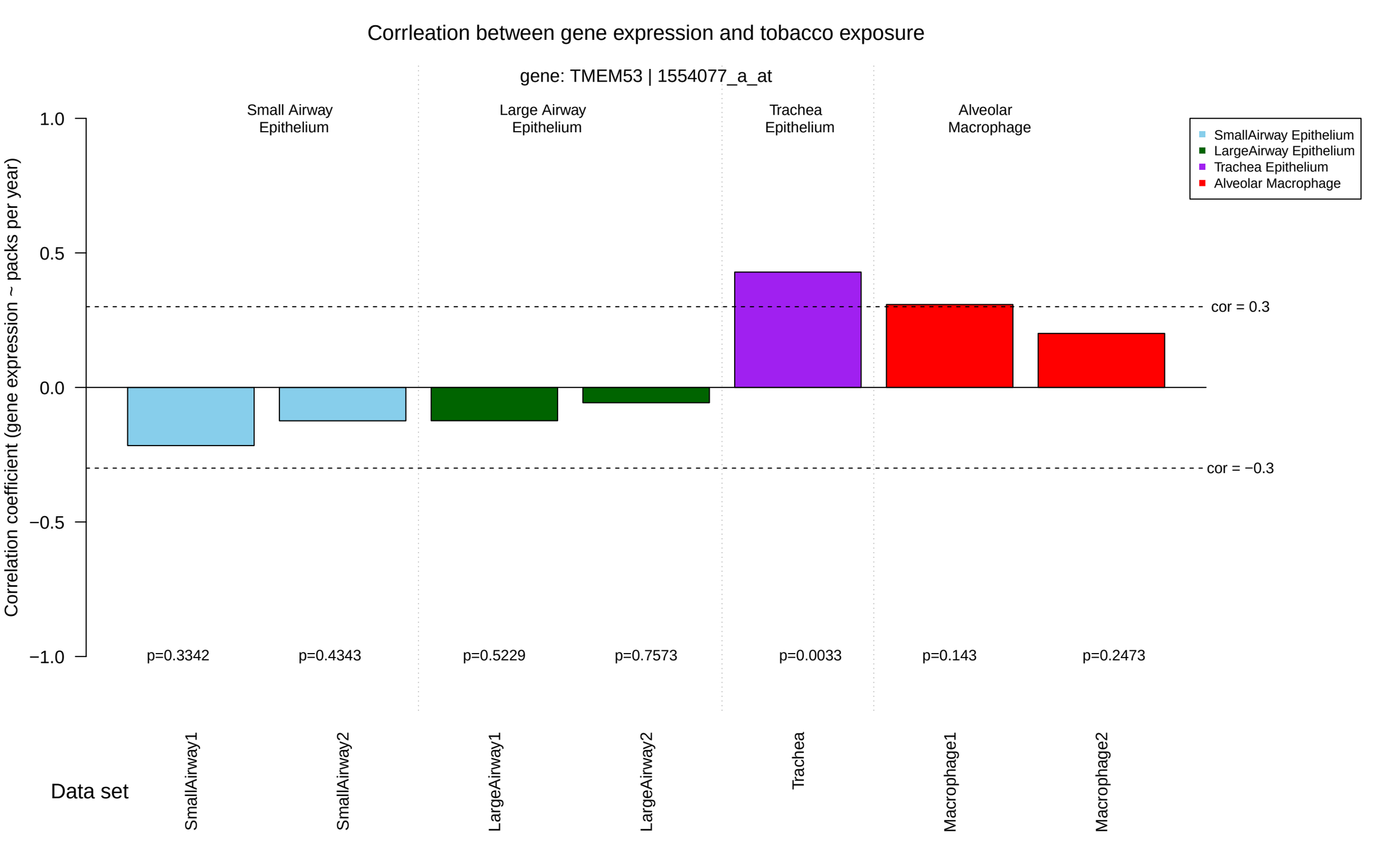

set: LargeAirway1 (p=0.4441)
TMEM53 | 1554077_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2546)
TMEM53 | 1554077_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0301)
TMEM53 | 1554077_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0659)
TMEM53 | 1554077_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0839)
TMEM53 | 1554077_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM53 | 1554077_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3342
p=0.4343
p=0.5229
p=0.7573
p=0.0033
p=0.143
p=0.2473
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2